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# EgoPed: Ego-Conditioned Pedestrian Body Motion Generation via Latent Diffusion

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**ABSTRACT** Simulation-based testing of automated vehicles needs pedestrians whose whole-body motion responds to the vehicle, yet existing motion generators condition on text, waypoints, goals or the static scene. We present EgoPed, a latent diffusion model that generates 3D full-body pedestrian motion directly from the ego vehicle's trajectory, encoded by a pretrained, frozen trajectory encoder into a single pooled token. On 11,786 paired sequences from nuScenes, Waymo and staged recordings of our own vehicle, split by scene, EgoPed follows the vehicle where a separately trained unconditional prior cannot: across three training seeds it lowers the root displacement error from 3.11 to 2.43 m and lifts the coupling between generated stopping decisions and the ground truth (Ego-Coupling Score) from 0.02 to 0.31; given another scene's vehicle, it falls back to the prior's level. The same experiments show that the Fréchet distance is blind to this coupling. It ranks our models almost in reverse of their per-condition accuracy, an ego-blind prior matches the best conditional model (3.29 vs. 3.24), and the two design choices it favours, full-sequence ego attention and interaction-weighted training data, both lower per-condition accuracy under a significance rule fixed before the test split was used. We therefore propose per-condition trajectory error and the Ego-Coupling Score, with scene-clustered statistics across training seeds, as the evaluation for this task. A reference-free audit of the pseudo-ground-truth poses shows that root trajectories are reliable while the body is a lightly articulated template, which bounds what any pose-level metric can mean on this data. Code, trained models and data will be released.

**INDEX TERMS** Pedestrian behaviour modelling, human motion synthesis, denoising diffusion models, autonomous driving, vehicle–pedestrian interaction, generative model evaluation.

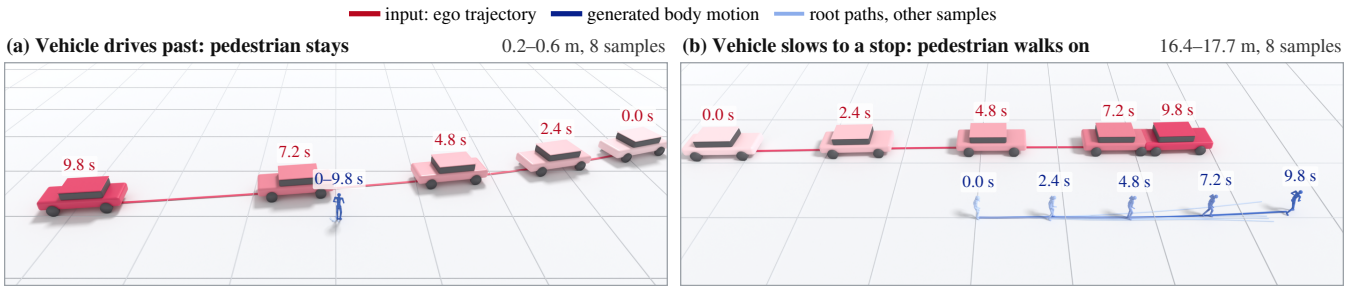



FIGURE 1: EgoPed generates full-body pedestrian motion from the ego vehicle's trajectory alone, given in the pedestrian's initial frame. Panels (a) and (b) share model and noise; only the input trajectory differs. Titles give the distance travelled across 8 noise seeds; time stamps pair vehicle and pedestrian; the grid spacing is 5 m; the vehicle shape is for scale only. Both conditions are from the validation partition and were chosen for illustration: no single manoeuvre decides stopping, and Section c measures the coupling over the whole test set.

## I. INTRODUCTION

GENERATING realistic pedestrian body motion is a core capability for automated-vehicle (AV) simulation. Simulation pipelines need diverse, plausible pedestrian behaviour to stress-test perception and planning, especially in the long tail of close-range encounters that are unsafe or impractical to collect on the road. That behaviour is coupled to the vehicle: its speed, path and proximity influence whether a pedestrian

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crosses, waits, accelerates or turns [1], and benchmarks such as JAAD [2] and PIE [3] annotate exactly that dependence. The models built on them, however, predict a discrete event or a ground-plane path [4]–[6], not an articulated body.

Full-body human motion synthesis has meanwhile advanced quickly through text-conditioned diffusion [7]–[10]. These generators are blind to the vehicle in exactly the way a simulator cannot afford: a person crossing a street is generated the same whether a car approaches at 50 km/h or stands still. Pedestrian-specific generators control the body through waypoints, goals or the static scene [11], [12], and concurrent work reaches the body only through an intermediate 2D path and a text prompt [13].

We present EgoPed, which generates the body motion directly from the ego vehicle's trajectory (Fig. 1). It builds on Motion Latent Diffusion (MLD) [7] and replaces the text condition with a pretrained, frozen trajectory encoder whose output is pooled into a single conditioning token. Showing that such a model *uses* the vehicle is harder than it looks, and most of this paper is about how to tell. The condition belongs to another agent and never determines the output, so the standard Fréchet distance (FID), which compares the marginal distribution of generated and real motion, can be excellent for a model that ignores the vehicle altogether. We therefore evaluate every model on per-condition root trajectory error and on an Ego-Coupling Score that measures whether generated stopping decisions follow the vehicle, with three training seeds for each main model, a test split that shares no scene with training, and a significance rule fixed before the test split was touched.

Our contributions are:

1) **Ego-conditioned body motion.** To our knowledge the first model that generates 3D pedestrian body motion directly from the ego-vehicle trajectory, where prior pedestrian generators condition on text, waypoints, goals or scene. Against a separately trained unconditional prior it lowers root trajectory error by 0.68 m and raises the Ego-Coupling Score from 0.02 to 0.31.
2) **Conditioning-aware evaluation.** Per-condition trajectory error and the Ego-Coupling Score, with scene-clustered statistics across seeds. With them we show that FID cannot see the conditioning: an ego-blind prior matches the best conditional FID, and both design choices FID prefers, full-sequence ego attention and interaction-weighted training data, lose on per-condition accuracy.
3) **Data and its limits.** 11,786 paired ego–pedestrian sequences from nuScenes [14], Waymo [15] and staged recordings of our research vehicle, split by scene, together with a reference-free audit showing that the pseudo-ground-truth root trajectories are reliable while the body poses are a sliding canonical template.

## II. RELATED WORK

### A. PEDESTRIAN BEHAVIOUR AND TRAJECTORY PREDICTION

Modelling how pedestrians behave around vehicles is a long-standing concern in intelligent transportation [1], [16]. Video benchmarks predict discrete events: JAAD [2] annotates crossing behaviour with its context and PIE [3] adds intention labels with ego-vehicle sensor readings, while Kotseruba *et al.* [17] show that reported gains depend strongly on evaluation design and the base rate of the predicted event. A second line predicts pedestrians as points on the ground plane, from social pooling [4], [18], [19] to heterogeneous agents and attention [5], [20], goal-conditioned prediction [21], [22] and diffusion [6], [23]; on the vehicle side MTR [24] and UniTraj [25] address multi-modal and cross-dataset prediction [26], [27]. We keep this literature's displacement metrics for the root of the body, and add an articulated body on top.

### B. HUMAN MOTION SYNTHESIS AND CONTROL

Generating 3D human motion progressed from action labels [28] to language [29], supported by HumanML3D and its 263-dimensional motion representation [30], which we adopt. Diffusion [31], [32] with classifier-free guidance [33] now dominates: MDM [8] and MotionDiffuse [9] denoise raw motion, while MLD [7] denoises in the latent space of a motion VAE [34], following latent diffusion for images [35]. Token-based [10], [36], [37] and retrieval-augmented [38] generators are alternatives. Spatially controlled generation, GMD [39] and OmniControl [40], constrains where the generated person's own body should be. Scene- [41]–[44] and partner-conditioned [45] models condition on static geometry or a second human. Our condition differs in kind: it is the trajectory of a different, independently moving agent that only shapes a distribution over plausible reactions. Frozen conditioning encoders are standard in image generation [35], [46], [47], and we test whether freezing matters here too.

### C. PEDESTRIAN MOTION GENERATION FOR DRIVING SCENES

Trace and Pace [11] combines a guided trajectory diffusion model, controlled through waypoints, speed and social groups, with a physics-based humanoid controller. PedGen [12] learns full-body pedestrian motion from web videos with noisy labels, using scene, goal and body shape as context. In both, control comes from these inputs rather than from the trajectory of a vehicle. Concurrent work, ControlPed [13], targets safety-critical vehicle–pedestrian scenarios in two stages: surrounding vehicles condition a 2D pedestrian trajectory, which a text-conditioned diffusion model lifts to body motion, so the body stage sees the vehicle only through that trajectory and the prompt. EgoPed generates the body motion directly from the vehicle trajectory. We compare against PedGen on the root trajectory in Section VI-F.

### D. EVALUATING CONDITIONAL GENERATIVE MODELS

Fréchet-distance scores compare generated and real samples through the statistics of a pretrained feature extractor [48]; the motion literature substitutes a motion encoder [30]. Their known failure modes concern how faithfully they measure the *marginal*: sample size and estimator bias [49], preprocessing [50], the extractor's own class structure [51], with kernel [52], embedding [53] and precision–recall [54] alternatives and related critiques of the Inception Score [55] and FVD [56]. The problem we document is orthogonal: a score can measure the marginal well and say nothing about whether the condition was used. Estimated body motion is also known to be physically imperfect [57]–[61]; we measure how imperfect our labels are, because it limits what a pose-level score can certify.

## III. METHOD

### A. PROBLEM FORMULATION

Given an ego-vehicle trajectory $\mathbf{e} \in \mathbb{R}^{T\times 2}$, the vehicle's ground-plane position over $T = 196$ frames at 20 fps expressed in the pedestrian's initial frame, we generate pedestrian body motion $\mathbf{x} \in \mathbb{R}^{T\times 263}$ in the HumanML3D representation [30], which encodes root velocity and rotation, joint positions and velocities and foot contacts. We model $p(\mathbf{x} \mid \mathbf{e})$ with a latent diffusion model (Fig. 2).

### B. MOTION VAE

We use the MLD VAE [7], a transformer encoder–decoder that compresses $T \times 263$ motion into $L = 4$ latent tokens of 256 dimensions. We train it from scratch on the training partition of our data (Section IV) for 6,000 epochs, with features normalized by that partition's statistics, and keep it frozen afterwards.

### C. EGO ENCODER

A lightweight transformer encodes the ego trajectory: a linear projection from 2 to 256 dimensions, sinusoidal positional encoding and four encoder layers with four heads give $\mathbf{c} \in \mathbb{R}^{196\times 256}$. EgoPed mean-pools $\mathbf{c}$ over time and applies a two-layer projection head, giving one conditioning token $\bar{\mathbf{c}} \in \mathbb{R}^{256}$, analogous to a pooled text embedding.

#### a: Pretraining.

Before diffusion training, the encoder is trained to regress the paired motion's latent: with $\bar{\mathbf{z}}$ the mean-pooled VAE posterior mean,

$$\mathcal{L}_{\text{pre}} = \|\bar{\mathbf{c}} - \bar{\mathbf{z}}\|^2 + \lambda\big(1 - \cos(\bar{\mathbf{c}}, \bar{\mathbf{z}})\big), \quad \lambda = 0.1. \tag{1}$$

The encoder is then *frozen* during diffusion training. Section VI-C tests both choices, together with a Full-seq variant that skips the pooling and lets the denoiser attend to all 196 tokens.

### D. DENOISER AND GUIDANCE

The MLD denoiser is a self-attention transformer (9 layers, 4 heads, 256 dimensions) over the concatenation of the noisy latent tokens, the timestep embedding and the condition; the first $L$ output tokens predict the noise:

$$\hat{\boldsymbol{\epsilon}} = \text{Transformer}\big([\mathbf{z}_t,\, \mathbf{t}_{\text{emb}},\, \bar{\mathbf{c}}]\big)_{1:L}. \tag{2}$$

It is trained with the DDPM noise-prediction loss $\mathbb{E}\,\|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}\|^2$ (1,000 steps) and sampled with 50 DDIM steps [32]. For classifier-free guidance [33], the ego input is zeroed for 10% of training samples, so the encoder's response to the all-zero trajectory is the unconditional embedding, and at inference

$$\hat{\boldsymbol{\epsilon}}_w = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \emptyset) + w\big(\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \bar{\mathbf{c}}) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \emptyset)\big). \tag{3}$$

## IV. DATA

### A. SOURCES AND PROCESSING

No dataset pairs vehicle trajectories with pedestrian body motion, so we built one from two complementary sources: the public driving datasets nuScenes [14] and Waymo [15], which supply scale and behavioural diversity, and a small set of deliberately *staged* high-interaction scenes recorded with our research vehicle AVA, which targets close-range encounters that are rare in public data. AVA, a Mercedes-Benz S-Class built up by the research group (Fig. 3), carries seven 128-layer LiDARs (Ouster OS0, OS1, OS2) on a roof rack, a raised mast and both bumpers, eight RGB cameras (Basler ACE2) facing forward, sideways and back, and a dual-antenna GNSS receiver (Novatel OEM7) for localization, which supplies the ego trajectory. The authors staged 17 interactive scenarios in Munich, such as crossing in front of the vehicle, stopping at the kerb, hesitating or turning around while the vehicle approaches, brakes or turns. The recordings are converted to the nuScenes format and annotated with 3D boxes and orientations using MS3D [62], [63].

For every annotated pedestrian in all three sources we estimate SMPL [64] poses with an adapted OmniRe [65], localizing the person in every camera through the projected 3D box and stitching the per-camera estimates into one sequence with the rig calibration. The poses are converted to the 263-dimensional HumanML3D representation [30], which retargets every sequence onto one canonical skeleton, so body shape is not represented. One pipeline resamples body motion and ego odometry to 20 fps and expresses both in a pedestrian-centric frame, translated and yaw-normalized to the pedestrian's first pose. Every sequence is also stored mirrored left to right. Sequences longer than 196 frames are windowed (about 40%), shorter ones are padded and masked.

### B. SCENE-LEVEL SPLIT

All partitions are split by scene: a scene, every pedestrian in it and every mirrored copy stay on one side (Table 1). This matters: in a file-level split of the same data, the mirrored twin of 83% of the held-out sequences was in the training set. Every number in this paper comes from models trained and evaluated on the scene-level split. We select models on the validation partition and evaluated the test partition once, after all selection was fixed. The staged AVA scenes contribute

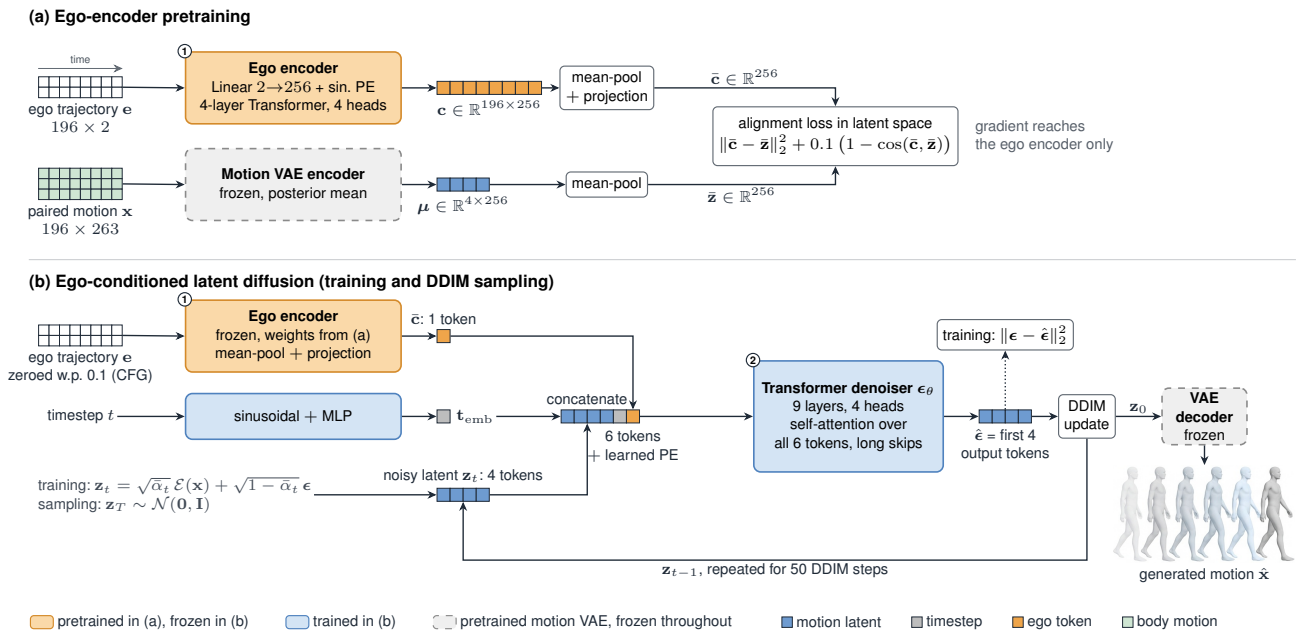



FIGURE 2: EgoPed architecture. **(a) Ego-encoder pretraining.** The ego encoder (1) is trained so that its mean-pooled, projected output $\bar{\mathbf{c}}$ matches the mean-pooled posterior mean $\bar{\mathbf{z}}$ of the frozen motion VAE for the paired motion. **(b) Latent diffusion.** The ego encoder is *frozen* and produces the single pooled token $\bar{\mathbf{c}}$; for classifier-free guidance the ego input is zeroed for 10% of training samples. At every denoising step the $L = 4$ noisy latent tokens $\mathbf{z}_t$, the timestep embedding and $\bar{\mathbf{c}}$ form one 6-token sequence for a self-attention transformer denoiser (2), whose first four output tokens predict the noise. Sampling runs 50 DDIM steps and decodes $\mathbf{z}_0$ with the frozen VAE decoder; only the denoiser is trained in (b). The Full-seq ablation skips the pooling and passes all 196 encoder tokens.



FIGURE 3: Our research vehicle AVA, used to record the staged high-interaction scenes. **Left:** top view of the roof-mounted sensor rack. **Right:** side view.

TABLE 1: Dataset composition: paired ego trajectory (196×2) and pedestrian body motion (196×263) per sequence, counted with mirrored copies. Partitions share no scene.

| Source | Train | Validation | Test | Total |
|---|---|---|---|---|
| AVA (ours, staged) | 22 | 6 | 6 | 34 |
| nuScenes | 3,594 | 472 | 448 | 4,514 |
| Waymo Open Dataset | 5,794 | 716 | 728 | 7,238 |
| Sequences | 9,410 | 1,194 | 1,182 | 11,786 |
| Scenes | 796 | 107 | 91 | 994 |

only six sequences to each held-out partition, so held-out results speak for nuScenes and Waymo.

*a: Interaction-aware data recipe (IA).*

As a data-side alternative we test a recipe that concentrates training on interactions. Each sequence gets an interaction score $s = s_{\text{travel}}\, s_{\text{prox}}\, s_{\text{bearing}}$ from the pedestrian's path length, the fraction of frames within 5 m of the vehicle and the swept relative bearing $(1 + \sum_t |\Delta\phi_t|/180°)$. IA samples training sequences with probability proportional to $s$ (floored at 0.01) and centres the 196-frame window on the closest approach, with $\pm 25\%$ jitter. Without IA, sequences are drawn uniformly and windows at random. Evaluation always uses one fixed window per sequence, identical for every model.

### C. POSE-LABEL AUDIT

The pose labels are estimates, and no reference such as mocap exists for these scenes. We therefore audit what can be checked without one, on all 13,958 extracted sequences long enough to score (Table 2). First, **body shape was never estimated**: every sequence shares one skeleton to within $10^{-5}$ relative bone length. Second, **the feet do not plant**: during locomotion one foot should be stationary, yet the slowest foot joint moves at a median 0.97 of root speed (0 = clean

TABLE 2: Reference-free audit of the pseudo-ground-truth poses (13,958 sequences, 10,900 with locomotion). Upper block: internal physical consistency. Lower block: our per-dimension standard deviation over HumanML3D's mocap statistics; values below 1 mean less variability than mocap.

| Internal consistency check | Result |
|---|---|
| Distinct skeletons among all sequences | 1 |
| Max. relative bone-length deviation | $6.3 \times 10^{-6}$ |
| Foot-skate ratio, median (0 = plant, 1 = slide) | 0.97 |
| AVA / nuScenes / Waymo | 0.73 / 0.99 / 0.95 |
| Never plant a foot (>5% of moving frames) | 94.6% |
| Near-static legs (ankle swing $<$ 5 cm) | 14.1% |
| Ground penetration $>$ 5 cm | 1.2% |
| **Variability vs. mocap (ours / HumanML3D std)** | **Ratio** |
| Root linear velocity | 1.29 |
| Joint rotations | 0.68 |
| Root-relative joint positions | 0.44 |
| Root rotational velocity | 0.37 |
| Root height | 0.26 |

footfalls, 1 = the body sliding as a rigid unit), and $94.6\%$ of moving sequences never hold a foot still for more than $5\%$ of their frames. Third, **articulation is damped**: against HumanML3D's mocap statistics, root translation varies more ($1.29\times$) but root-relative joint positions, root height and turning far less ($0.44\times$, $0.26\times$, $0.37\times$); part of that gap is content, since our data is people walking near roads. In an earlier version of this pipeline, a VAE round-trip of the labels and samples of trained models slid just as much (0.93–0.94), so the sliding lives in the labels and the latent space, not in the generator. A pilot that re-solves the legs against the measured root removes the sliding (skate 0.15 after fine-tuning the VAE on 1,418 corrected sequences), so relabelling is viable, but it changes the target distribution and is left to future work.

The consequence for evaluation is direct. Root trajectories are well tracked; the body is a lightly articulated canonical template sliding along them. Our conditioning evidence, trajectory error and stopping decisions, is computed from the root. FID scores the body against this damped target and should be read as a within-benchmark comparison only.

## V. EVALUATION PROTOCOL

*a: Distributional metrics.*

FID, Diversity and MultiModality (MM) follow the HumanML3D protocol [30], with its pretrained motion encoders as feature extractor: FID compares generated and real feature distributions, Diversity is the mean pairwise distance across generated samples (closer to the ground truth's 5.26–5.40 is better), and MM the mean pairwise distance among 10 samples of one condition. Each is the mean of three generation passes. We do not report R-Precision: it retrieves with each model's own ego encoder, so it is not comparable across encoders, and it falls to chance for a trainable encoder.

*b: Per-condition metrics.*

For each test condition we draw $K{=}5$ samples, take the pelvis ground-plane path and compare it with the recorded pedestrian over the true sequence length: ADE and FDE average over the samples, minADE and minFDE take the best of five. Mean ADE rewards samples that concentrate on the likely outcome, min-of-$K$ rewards coverage, so we read them together.

*c: Ego-Coupling Score.*

A pedestrian *stops* if the root speed stays below 0.2 m/s for at least 1 s. For condition $\mathbf{e}$, $\hat{p}(\text{stop} \mid \mathbf{e})$ is the fraction of the $K$ samples that stop. The Ego-Coupling Score (ECS) is the mean of $\hat{p}$ over conditions whose recorded pedestrian stops minus its mean over conditions whose pedestrian walks: 0 for a model that ignores the vehicle, 1 for perfect coupling. We also report the Brier score of $\hat{p}$ against the recorded label. Stopping occurs in 24% of test conditions; crossing the vehicle's path occurs in about 2% and is too rare to score.

*d: Seeds, statistics and selection.*

Every model is trained with three seeds. Each seed's checkpoint is the one with the lowest validation FID among 50 saved every 100 epochs, and each model's guidance scale is the one in $\{5, 10, 15\}$ with the lowest validation FID averaged over its seeds. Unconditional models see the real ego trajectory during training-time validation, so their checkpoints are chosen by an offline validation pass with the ego zeroed. Contrasts average a model's seeds and resample scenes (2,000 scene-clustered bootstrap draws), because pedestrians of one scene are not independent. Before the test partition was used, we fixed one rule: a model beats another only if the 95% interval excludes zero *and* all nine seed pairs agree in sign. The bootstrap captures sampling of scenes but not of seeds; the second condition covers that. The primary contrast is EgoPed against Full-seq on mean ADE.

## VI. RESULTS

### A. EgoPed USES THE VEHICLE

Table 3 gives the test results and Table 4 the contrasts. Against a prior trained on the same data without the ego trajectory, EgoPed lowers mean ADE by 0.68 m (3.11 to 2.43 m) and FDE by 1.27 m, and raises the Ego-Coupling Score from 0.02 to 0.31, with every seed pair agreeing. The prior is better on minADE, as any sampler that spreads its five samples widely must be; minADE alone is no test of conditioning, and the prior's ECS of 0.02 shows it ignores the vehicle, as it was built to. The coupling is the same on validation (ECS 0.36 against 0.02) and holds on both large sources: on the nuScenes and Waymo test sequences EgoPed reaches ECS 0.32 and 0.26, the prior $-0.01$ and 0.04. It is specific to the

TABLE 3: Test partition (1,182 conditions, 91 scenes), each model the mean over three seeds with the seed range in brackets; single-seed rows have no range. FID and Diversity use the HumanML3D evaluator; ADE to ECS are per-condition and computed from $K=5$ samples. Ground-truth Diversity is 5.26–5.40. EgoPed is the pooled frozen encoder. Bold: best of the four main models per column.

| Model | FID ↓ | Div. | ADE ↓ | FDE ↓ | minADE ↓ | minFDE ↓ | ECS ↑ | Brier ↓ |
|---|---|---|---|---|---|---|---|---|
| EgoPed | 5.67 [5.36, 6.26] | 5.63 | **2.43** [2.38, 2.48] | **5.11** | 1.65 [1.59, 1.74] | 3.40 | **0.31** [0.29, 0.34] | 0.191 |
| Full-seq | 4.16 [2.96, 4.91] | 5.78 | 2.69 [2.52, 2.98] | 5.67 | **1.54** [1.49, 1.57] | **3.18** | 0.27 [0.24, 0.29] | **0.178** |
| EgoPed + IA | 4.84 [4.69, 5.04] | 5.83 | 2.93 [2.86, 3.03] | 6.23 | 2.04 [1.98, 2.13] | 4.29 | 0.21 [0.18, 0.25] | 0.195 |
| Full-seq + IA | **3.24** [2.73, 4.14] | 5.89 | 3.01 [2.82, 3.14] | 6.50 | 2.07 [1.72, 2.31] | 4.42 | 0.10 [0.06, 0.12] | 0.214 |
| Full-seq, encoder trainable (1 seed) | 6.54 | 5.32 | 2.67 | 5.55 | 1.39 | 2.79 | 0.04 | 0.209 |
| Trained prior (no ego, 1 seed) | 4.67 | 5.79 | 3.11 | 6.38 | 1.31 | 2.64 | 0.02 | 0.214 |
| Trained prior + IA (1 seed) | 3.29 | 5.47 | 3.00 | 6.30 | 1.52 | 3.09 | 0.01 | 0.220 |

TABLE 4: Contrasts on the test partition: difference of seed means, positive when the first model is better, with 95% scene-bootstrap interval. * meets the rule fixed before the test pass (interval excludes 0 and all seed pairs agree in sign). Rows 1–2 compare encoders at a matched data recipe, rows 3–4 the data recipe per encoder, rows 5–8 each model against the prior trained on its data recipe.

| Contrast | ADE (m) | minADE (m) | ECS |
|---|---|---|---|
| EgoPed vs. Full-seq | $+0.26^{*}$ [0.17, 0.33] | $-0.11^{*}$ [-0.18, -0.05] | $+0.04$ [-0.00, 0.08] |
| EgoPed +IA vs. Full-seq +IA | $+0.08$ [-0.04, 0.20] | $+0.03$ [-0.08, 0.15] | $+0.11^{*}$ [0.05, 0.18] |
| EgoPed vs. EgoPed +IA | $+0.50^{*}$ [0.30, 0.70] | $+0.39^{*}$ [0.20, 0.57] | $+0.09^{*}$ [0.02, 0.16] |
| Full-seq vs. Full-seq +IA | $+0.31$ [0.11, 0.55] | $+0.53^{*}$ [0.34, 0.74] | $+0.17^{*}$ [0.09, 0.25] |
| EgoPed vs. prior | $+0.68^{*}$ [0.49, 0.86] | $-0.34^{*}$ [-0.46, -0.22] | $+0.29^{*}$ [0.19, 0.39] |
| Full-seq vs. prior | $+0.42^{*}$ [0.22, 0.63] | $-0.23^{*}$ [-0.34, -0.12] | $+0.26^{*}$ [0.16, 0.34] |
| EgoPed +IA vs. prior+IA | $+0.07$ [-0.23, 0.34] | $-0.52^{*}$ [-0.77, -0.28] | $+0.21^{*}$ [0.13, 0.30] |
| Full-seq +IA vs. prior+IA | $-0.01$ [-0.29, 0.25] | $-0.55^{*}$ [-0.80, -0.33] | $+0.09^{*}$ [0.04, 0.15] |

vehicle in the scene: given another scene's vehicle, EgoPed falls to the prior's level (Section VI-D).

### B. FID DOES NOT SEE THE CONDITIONING

Read by FID, Table 3 tells a different story. The best FID belongs to Full-seq +IA (3.24), with the ego-blind prior trained on the same recipe level with it (3.29), and EgoPed has the worst FID of the four conditional cells (5.67) while being the most accurate on mean ADE and the most strongly coupled. Across all 15 training runs, the rank correlation of FID with mean ADE is $-0.74$ and with ECS $+0.48$: the lower a run's FID, the larger its trajectory error and the weaker its coupling (Fig. 4). Interaction-weighted data illustrates the mechanism: it lowers FID for the conditional model and the prior alike, because it changes which motions the model sees, and at the same time it lowers every per-condition metric for both encoders (Table 4, rows 3–4); the IA models are no better than their own prior on mean ADE. A metric of the marginal can reward a change that removes conditioning. We therefore select checkpoints by FID, as is standard, but rank models only by the per-condition metrics.

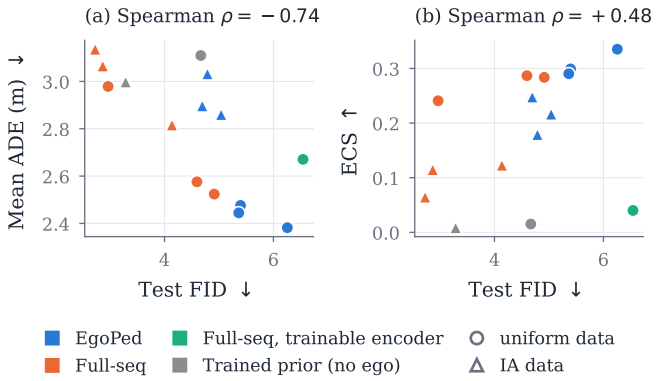



FIGURE 4: Test FID against per-condition accuracy, one point per training run (15 runs; colour: model, marker: data recipe). Lower FID goes with *larger* trajectory error (a) and *weaker* coupling (b); the ego-blind prior on IA data sits among the best FIDs.

### C. ENCODER DESIGN

*a: Pooled against full-sequence conditioning.*

Letting the denoiser attend to all 196 encoder tokens instead of one pooled token makes per-condition predictions worse on average: Full-seq trails EgoPed by 0.26 m in mean ADE (Table 4), the pre-specified primary contrast, and did so on validation as well (0.37 m). The trade-off is spread: Full-seq's samples cover more outcomes (MM 3.92 vs. 3.10, better minADE by 0.11 m), and its FID is lower. On coupling the two encoders are close without IA (ECS 0.27 vs. 0.31, below our rule on test, met on validation), and EgoPed couples clearly better with IA (0.21 vs. 0.10). We attribute part of the gap to pretraining: the alignment objective supervises only the pooled token, so the pooled encoder hands the denoiser exactly the representation it was trained to produce, whereas the 196 per-frame tokens were never aligned individually.

*b: Freezing and pretraining.*

Table 5 trains EgoPed's encoder together with the denoiser, initialised once from pretraining and once from scratch, with three seeds each and the same selection as every other model. Freezing matters: the frozen encoder beats the trainable pretrained one on all four per-condition metrics under our rule (ADE by 0.09 m, minADE by 0.25 m, ECS by 0.07, Brier by 0.025), as it did on validation for all but ADE. FID does not separate the two (5.67 vs. 5.65). Pretraining by itself does not help: once the encoder is trainable, starting from the pretrained weights gives the same result as starting from scratch on every metric and on both partitions. The pretrained representation thus helps only as long as diffusion training cannot change it. The trainable encoders also concentrate their samples (MM 2.0–2.1 against 3.10), which costs them minADE without improving the mean. Validation FID selected early checkpoints for four of the six trainable runs (epochs 999–1099). Because their encoders might still have been learning, we decided before the test pass to also report each run's last checkpoint; it shows the same picture with worse FID, and the ADE gap to the frozen encoder (0.07 m) no longer meets the rule. The pooled encoders learn the coupling even when trainable; a single-seed run of Full-seq with a trainable encoder did not: its alignment between encoder output and motion latent stays at chance through all 5,000 epochs, and the selected model is close to ego-blind (ECS 0.04 against 0.27 for the frozen Full-seq, Table 3).

TABLE 5: Encoder ablations of EgoPed on the test partition, means over three seeds. Upper rows: checkpoint and guidance selected on validation like every other model (guidance 15, 15, 10). Lower rows: the last checkpoint of the same runs at the same guidance, reported as decided before the test pass.

| Encoder | FID ↓ | MM | ADE ↓ | minADE ↓ | ECS ↑ |
|---|---|---|---|---|---|
| Pretrained, frozen (EgoPed) | 5.67 | 3.10 | 2.43 | 1.65 | 0.31 |
| Pretrained, trainable | 5.65 | 2.15 | 2.53 | 1.90 | 0.23 |
| From scratch, trainable | 5.77 | 2.03 | 2.50 | 1.87 | 0.24 |
| *Last checkpoint (epoch 4,999)* | | | | | |
| Pretrained, trainable | 6.78 | 1.65 | 2.51 | 2.01 | 0.23 |
| From scratch, trainable | 7.03 | 1.74 | 2.50 | 1.97 | 0.23 |

TABLE 6: Ego controls for EgoPed on the test partition (three seeds at their selected checkpoints, no retraining). Horizon rows show the first $S$ seconds of the ego trajectory and continue it at the velocity of the last 0.5 s. The prior row is the separately trained model for reference.

| Ego input | ADE ↓ | minADE ↓ | ECS ↑ | Brier ↓ |
|---|---|---|---|---|
| Full trajectory (9.8 s) | 2.43 | 1.65 | 0.31 | 0.191 |
| First 4 s, then constant velocity | 2.50 | 1.72 | 0.32 | 0.191 |
| First 2 s, then constant velocity | 2.52 | 1.74 | 0.33 | 0.194 |
| First 1 s, then constant velocity | 2.60 | 1.81 | 0.31 | 0.205 |
| Vehicle stationary at its start | 3.04 | 2.07 | 0.06 | 0.239 |
| Vehicle from another scene | 3.09 | 2.31 | 0.00 | 0.312 |
| Zeroed (unconditional branch) | 3.03 | 1.31 | 0.03 | 0.210 |
| Trained prior (no ego) | 3.11 | 1.31 | 0.02 | 0.214 |

### D. WHICH PART OF THE TRAJECTORY IS USED

Table 6 feeds EgoPed's three seeds manipulated ego input, without retraining. Given the trajectory of a vehicle from another scene, EgoPed is no better than the trained prior (ADE 3.09 against 3.11 m) and its coupling vanishes (ECS 0.00); its Brier score (0.31) is far worse than the prior's, because it reacts confidently to the wrong vehicle. Zeroing the input gives the prior's numbers too. A vehicle that stays at its start position removes most of the coupling as well (ECS 0.06), so the model reacts to how the vehicle moves, not only to where it is. Showing only the first second and continuing it at constant velocity keeps the coupling (ECS 0.31; 0.30 against 0.36 on validation, not a difference under our rule) and most of the gain in ADE (2.60 m, between 2.43 with the full trajectory and 3.04 with a stationary vehicle). The stop decision is thus carried by the vehicle's early motion, and its later trajectory refines the path: the ADE cost of the 1 s horizon meets our rule on both partitions. All differences between the full input and the stationary, foreign or zeroed vehicle meet it as well.

### E. GUIDANCE

Guidance trades coverage for FID without changing coupling (Table 7, validation). For both encoders a higher scale lowers FID but raises ADE and minADE, while ECS stays flat. Selecting the scale by FID, as our protocol does, therefore picks the least accurate of the three settings; at $w{=}5$, EgoPed's validation ADE would be 0.03 m lower and its minADE 0.21 m lower. We keep the pre-specified selection and report the trade-off.

TABLE 7: Guidance scale on the validation partition, mean over three seeds at each seed's selected checkpoint. Bold: the scale selected by validation FID.

| Model | $w$ | FID ↓ | ADE ↓ | minADE ↓ | ECS ↑ |
|---|---|---|---|---|---|
| EgoPed | 5 | 4.61 | 2.24 | 1.22 | 0.37 |
| | 10 | 4.07 | 2.25 | 1.36 | 0.37 |
| | **15** | 3.90 | 2.27 | 1.43 | 0.36 |
| Full-seq | 5 | 3.29 | 2.46 | 1.23 | 0.30 |
| | 10 | 2.80 | 2.55 | 1.33 | 0.30 |
| | **15** | 2.74 | 2.64 | 1.38 | 0.30 |

TABLE 8: First 1.95 s of the root path on the test partition (meters, $K$=5). PedGen samples come from its context-free model; it was trained on other data and never sees the vehicle.

| Model | ADE ↓ | FDE ↓ | minADE ↓ | minFDE ↓ |
|---|---|---|---|---|
| EgoPed (3 seeds) | **0.575** | **1.157** | 0.370 | 0.744 |
| Full-seq (3 seeds) | 0.632 | 1.278 | 0.341 | 0.686 |
| Trained prior | 0.750 | 1.514 | **0.286** | **0.572** |
| PedGen, no context [12] | 0.794 | 1.570 | 0.423 | 0.814 |

### F. COMPARISON WITH PEDGEN

PedGen [12] generates pedestrian motion from scene, goal and body-shape context and is trained on web video. Its released context-free model cannot take a vehicle trajectory, so we use it as an ego-blind reference on the one quantity both models share: the root path over its 2-second horizon, compared over the first 1.95 s of our test conditions with every path starting at the origin and facing forward (Table 8). EgoPed is 0.22 m closer on mean ADE (interval excluding zero for every seed). Our own trained prior is closer than PedGen as well, so part of the gap is the domain: PedGen's samples walk at a median 0.65 m/s against 1.26 m/s in our data. The comparison shows that a published generator does not transfer to this setting without retraining, not that one architecture is better than the other.

### G. QUALITATIVE RESULTS

Fig. 1 shows one EgoPed model and one noise seed under two vehicle trajectories, and Fig. 5 shows generated motion in a simulator. For visualisation, the generated HumanML3D joint sequences are retargeted onto CARLA pedestrian skeletons and replayed in a CARLA town together with an ego vehicle that follows the conditioning trajectory [Mohan: please describe the retargeting and scene placement, CARLA version and town briefly].

## VII. DISCUSSION

*a: What the evaluation changes.*

Had we ranked by FID, this paper would recommend full-sequence attention with interaction-weighted data, the combination with the weakest coupling to the vehicle among our conditional models. What prevents that conclusion is the combination of a split that shares no scene, several seeds with a rule that requires every seed pair to agree, and per-condition metrics that a model ignoring the vehicle cannot pass. We suggest the same minimum for conditional motion generation in general: report a separately trained unconditional prior on the same data recipe, and report at least one per-condition metric next to FID.

*b: A trade-off, not a free choice.*

EgoPed concentrates its samples near the likely outcome, Full-seq spreads them. For a simulator that must cover rare reactions, Full-seq's higher MM and better minADE may be worth its higher mean error; for one that must reproduce typical reactions to a given manoeuvre, EgoPed is the better choice. Both are clearly conditioned, and the pooled design is cheaper (6 tokens per denoising step instead of 201).

*c: Outlook: spill-over to robotics.*

Nothing in the model is specific to cars: its input is the 2D trajectory of a moving agent in the pedestrian's frame. Simulating how people respond to delivery or service robots is a natural next use, and datasets recorded from mobile platforms such as JRDB [67] exist. This remains untested: we have no robot data, and robots differ from cars in size, speed and how people read them, so we expect retraining to be necessary.

*d: Broader impact.*

The intended use is simulation-based safety testing, which avoids exposing real pedestrians to risk. Generated motion inherits the demographics and settings of its sources, U.S. and European cities, and should be validated before use elsewhere; it contains no personally identifiable information.

## VIII. LIMITATIONS

- **Body pose quality.** The labels are estimated: one canonical skeleton, feet that do not plant and damped articulation (Section IV-C). Root trajectories are reliable, the body is not. Relabelling, for instance with the contact correction we piloted, and retraining is the largest open item.
- **Known future.** EgoPed is trained on the vehicle's whole 9.8 s trajectory, which suits simulation, where the ego plan is given, but not online prediction. Its stop decisions survive when only the first second is shown and continued at constant velocity (Section VI-D), but a model trained on past ego motion only is untested.
- **Condition and coverage.** The condition is the ego trajectory only, without map, other agents or traffic control. Held-out results rest on nuScenes and Waymo, since the staged AVA scenes contribute six held-out sequences per partition.
- **Metrics.** ECS covers one decision, stopping; crossing is too rare in our windows to score. FID uses an evaluator

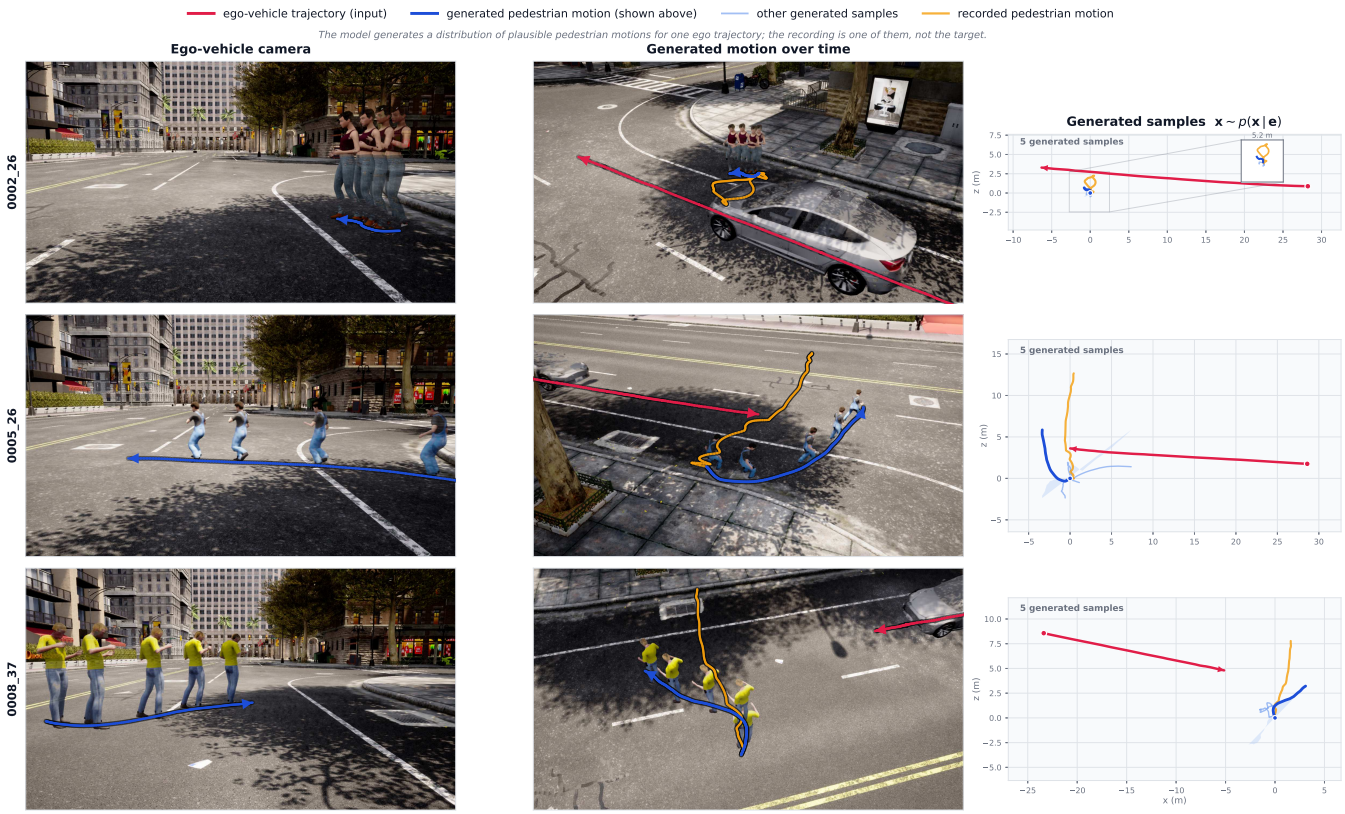



FIGURE 5: To be regenerated from EgoPed on test-partition conditions, including Waymo, AVA and a failure case; the current panels come from an earlier full-sequence model. Generated body motion retargeted onto pedestrian characters in the CARLA simulator [66], one ego condition per row. **Left:** view from the ego vehicle. **Middle:** third-person view over time with the ego trajectory (red), the shown sample's root path (dark blue) and the recorded pedestrian (orange). **Right:** bird's-eye view of five samples for the same ego trajectory. The recording is one outcome of the distribution, not its target, and the feet slide as in the labels (Section IV-C).

trained on mocap, so absolute values compare only within this benchmark, and we report no perceptual study.

## IX. CONCLUSION

EgoPed generates full-body pedestrian motion from the ego vehicle's trajectory with a pretrained, frozen, pooled trajectory encoder and latent diffusion. Against a prior trained without the vehicle, it lowers trajectory error by 0.68 m and lifts the coupling of generated stopping decisions from 0.02 to 0.31, across three seeds on a test split that shares no scene with training. The same experiments show that FID cannot certify such a model: it favours the design choices that weaken the coupling and places an ego-blind prior level with the best conditional model. Per-condition trajectory error and the Ego-Coupling Score, with scene-level statistics over several seeds, are what we propose for evaluating generation conditioned on another agent.

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## REFERENCES

[1] F. Camara, N. Bellotto, S. Cosar, F. Weber, D. Nathanael, M. Althoff, J. Wu, J. Ruenz, A. Dietrich, G. Markkula, A. Schieben, F. Tango, N. Merat, and C. Fox, "Pedestrian Models for Autonomous Driving Part II: High-Level Models of Human Behavior," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, pp. 5453–5472, 2021.

[2] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, "Are They Going to Cross? A Benchmark Dataset and Baseline for Pedestrian Crosswalk Behavior," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2017, pp. 206–213.

[3] A. Rasouli, I. Kotseruba, T. Kunic, and J. Tsotsos, "PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 6261–6270.

[4] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human Trajectory Prediction in Crowded Spaces," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 961–971.

[5] T. Salzmann, B. Ivanovic, P. Chakravarty, and M. Pavone, *Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data*, 2020, pp. 683–700.

[6] T. Gu, G. Chen, J. Li, C. Lin, Y. Rao, J. Zhou, and J. Lu, "Stochastic Trajectory Prediction via Motion Indeterminacy Diffusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 17 092–17 101.

[7] X. Chen, B. Jiang, W. Liu, Z. Huang, B. Fu, T. Chen, and G. Yu, "Executing your commands via motion diffusion in latent space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023.

[8] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-Or, and A. H. Bermano, "Human motion diffusion model," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[9] M. Zhang, Z. Cai, L. Pan, F. Hong, X. Guo, L. Yang, and Z. Liu, "MotionDiffuse: Text-driven human motion generation with diffusion model," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2024.

[10] C. Guo, Y. Mu, M. G. Javed, S. Wang, and L. Cheng, "MoMask: Generative masked modeling of 3d human motions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024.

[11] D. Rempe, Z. Luo, X. B. Peng, Y. Yuan, K. Kitani, K. Kreis, S. Fidler, and O. Litany, "Trace and pace: Controllable pedestrian animation via guided trajectory diffusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 13 756–13 766.

[12] Z. Liu, J. Lin, W. Wu, and B. Zhou, "Learning to generate diverse pedestrian movements from web videos with noisy labels," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[13] S. Liu, M. Li, H. Yu, H. Lin, Q. Zhou, B. Nie, and D. Zhao, "Controllable and photorealistic pedestrian risky motion generation for end-to-end driving safety evaluation," *arXiv preprint arXiv:2610.06171*, 2026.

[14] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A Multimodal Dataset for Autonomous Driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11 618–11 628.

[15] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine, V. Vasudevan, W. Han, J. Ngiam, H. Zhao, A. Timofeev, S. Ettinger, M. Krivokon, A. Gao, A. Joshi, Y. Zhang, J. Shlens, Z. Chen, and D. Anguelov, "Scalability in Perception for Autonomous Driving: Waymo Open Dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 2443–2451.

[16] F. Camara, N. Bellotto, S. Cosar, D. Nathanael, M. Althoff, J. Wu, J. Ruenz, A. Dietrich, and C. W. Fox, "Pedestrian Models for Autonomous Driving Part I: Low-Level Models, From Sensing to Tracking," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, pp. 6131–6151, 2021.

[17] I. Kotseruba, A. Rasouli, and J. K. Tsotsos, "Benchmark for Evaluating Pedestrian Action Prediction," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2021, pp. 1257–1267.

[18] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, and A. Alahi, "Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 2255–2264.

[19] A. Mohamed, K. Qian, M. Elhoseiny, and C. Claudel, "Social-STGCNN: A Social Spatio-Temporal Graph Convolutional Neural Network for Human Trajectory Prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 14 412–14 420.

[20] Y. Yuan, X. Weng, Y. Ou, and K. Kitani, "AgentFormer: Agent-Aware Transformers for Socio-Temporal Multi-Agent Forecasting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 9793–9803.

[21] K. Mangalam, H. Girase, S. Agarwal, K.-H. Lee, E. Adeli, J. Malik, and A. Gaidon, *It Is Not the Journey But the Destination: Endpoint Conditioned Trajectory Prediction*, 2020, pp. 759–776.

[22] K. Mangalam, Y. An, H. Girase, and J. Malik, "From Goals, Waypoints & Paths To Long Term Human Trajectory Forecasting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 15 213–15 222.

[23] W. Mao, C. Xu, Q. Zhu, S. Chen, and Y. Wang, "Leapfrog diffusion model for stochastic trajectory prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023.

[24] S. Shi, L. Jiang, D. Dai, and B. Schiele, "Motion Transformer with Global Intention Localization and Local Movement Refinement," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 6531–6543.

[25] L. Feng, M. Bahari, K. M. B. Amor, É. Zablocki, M. Cord, and A. Alahi, "UniTraj: A unified framework for scalable vehicle trajectory prediction," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.

[26] A. Rudenko, L. Palmieri, M. Herman, K. M. Kitani, D. M. Gavrila, and K. O. Arras, "Human Motion Trajectory Prediction: A Survey," *Int. J. Robot. Res.*, 2020.

[27] J. Liu, X. Mao, Y. Fang, D. Zhu, and M. Q.-H. Meng, "A Survey on Deep-Learning Approaches for Vehicle Trajectory Prediction in Autonomous Driving," in *Proc. IEEE Int. Conf. Robot. Biomimetics (ROBIO)*, 2021, pp. 978–985.

[28] M. Petrovich, M. J. Black, and G. Varol, "Action-Conditioned 3D Human Motion Synthesis with Transformer VAE," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 10 965–10 975.

[29] ——, *TEMOS: Generating Diverse Human Motions from Textual Descriptions*, 2022, pp. 480–497.

[30] C. Guo, S. Zou, X. Zuo, S. Wang, W. Ji, X. Li, and L. Cheng, "Generating diverse and natural 3d human motions from text," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022.

[31] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020.

[32] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[33] J. Ho and T. Salimans, "Classifier-free diffusion guidance," *arXiv preprint arXiv:2207.12598*, 2022.

[34] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014.

[35] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022.

[36] J. Zhang, Y. Zhang, X. Cun, Y. Zhang, H. Zhao, H. Lu, X. Shen, and Y. Shan, "T2M-GPT: Generating human motion from textual descriptions with discrete representations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023.

[37] B. Jiang, X. Chen, W. Liu, J. Yu, G. Yu, and T. Chen, "MotionGPT: Human Motion as a Foreign Language," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, pp. 20 067–20 079.

[38] M. Zhang, X. Guo, L. Pan, Z. Cai, F. Hong, H. Li, L. Yang, and Z. Liu, "ReMoDiffuse: Retrieval-augmented motion diffusion model," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023.

[39] K. Karunratanakul, K. Preechakul, S. Suwajanakorn, and S. Tang, "Guided motion diffusion for controllable human motion synthesis," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023.

[40] Y. Xie, V. Jampani, L. Zhong, D. Sun, and H. Jiang, "OmniControl: Control Any Joint at Any Time for Human Motion Generation," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[41] M. Hassan, V. Choutas, D. Tzionas, and M. Black, "Resolving 3D Human Pose Ambiguities With 3D Scene Constraints," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 2282–2292.

[42] M. Hassan, D. Ceylan, R. Villegas, J. Saito, J. Yang, Y. Zhou, and M. Black, "Stochastic Scene-Aware Motion Prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 11 354–11 364.

[43] Z. Wang, Y. Chen, T. Liu, Y. Zhu, W. Liang, and S. Huang, "HUMANISE: Language-Conditioned Human Motion Generation in 3D Scenes," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 14 959–14 971.

[44] N. Jiang, Z. Zhang, H. Li, X. Ma, Z. Wang, Y. Chen, T. Liu, Y. Zhu, and S. Huang, "Scaling Up Dynamic Human-Scene Interaction Modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 1737–1747.

[45] H. Liang, W. Zhang, W. Li, J. Yu, and L. Xu, "InterGen: Diffusion-based multi-human motion generation under complex interactions," *Int. J. Comput. Vis.*, 2024.

[46] H. Ye, J. Zhang, S. Liu, X. Han, and W. Yang, "IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models," *arXiv preprint arXiv:2308.06721*, 2023.

[47] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023.

[48] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

[49] M. J. Chong and D. Forsyth, "Effectively Unbiased FID and Inception Score and Where to Find Them," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 6069–6078.
[50] G. Parmar, R. Zhang, and J.-Y. Zhu, "On Aliased Resizing and Surprising Subtleties in GAN Evaluation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 11 400–11 410.
[51] T. Kynkäänniemi, T. Karras, M. Aittala, T. Aila, and J. Lehtinen, "The Role of ImageNet Classes in Fréchet Inception Distance," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.
[52] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, "Demystifying MMD GANs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.
[53] S. Jayasumana, S. Ramalingam, A. Veit, D. Glasner, A. Chakrabarti, and S. Kumar, "Rethinking FID: Towards a Better Evaluation Metric for Image Generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 9307–9315.
[54] T. Kynkäänniemi, T. Karras, S. Laine, J. Lehtinen, and T. Aila, "Improved Precision and Recall Metric for Assessing Generative Models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.
[55] S. Barratt and R. Sharma, "A Note on the Inception Score," *arXiv preprint arXiv:1801.01973*, 2018.
[56] T. Unterthiner, S. van Steenkiste, K. Kurach, R. Marinier, M. Michalski, and S. Gelly, "Towards Accurate Generative Models of Video: A New Metric & Challenges," *arXiv preprint arXiv:1812.01717*, 2018.
[57] D. Rempe, L. J. Guibas, A. Hertzmann, B. Russell, R. Villegas, and J. Yang, *Contact and Human Dynamics from Monocular Video*, 2020, pp. 71–87.
[58] K. Xie, T. Wang, U. Iqbal, Y. Guo, S. Fidler, and F. Shkurti, "Physics-based Human Motion Estimation and Synthesis from Videos," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 11 512–11 521.
[59] S. Zhang, Y. Zhang, F. Bogo, M. Pollefeys, and S. Tang, "Learning Motion Priors for 4D Human Body Capture in 3D Scenes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 11 323–11 333.
[60] Y. Yuan, J. Song, U. Iqbal, A. Vahdat, and J. Kautz, "PhysDiff: Physics-Guided Human Motion Diffusion Model," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 15 964–15 975.
[61] L. Mourot, L. Hoyet, F. L. Clerc, and P. Hellier, "UnderPressure: Deep Learning for Foot Contact Detection, Ground Reaction Force Estimation and Footskate Cleanup," *Comput. Graph. Forum*, vol. 41, pp. 195–206, 2022.
[62] D. Tsai, J. S. Berrio, M. Shan, E. Nebot, and S. Worrall, "Ms3d++: Ensemble of experts for multi-source unsupervised domain adaptation in 3d object detection," *IEEE Trans. Intell. Veh.*, pp. 1–16, 2024.
[63] ——, "Ms3d: Leveraging multiple detectors for unsupervised domain adaptation in 3d object detection," in *Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC)*, 2023, pp. 140–147.
[64] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, "SMPL: A skinned multi-person linear model," *ACM Trans. Graph.*, vol. 34, no. 6, pp. 248:1–248:16, Oct. 2015.
[65] Z. Chen, J. Yang, J. Huang, R. de Lutio, J. M. Esturo, B. Ivanovic, O. Litany, Z. Gojcic, S. Fidler, M. Pavone, L. Song, and Y. Wang, "Omnire: Omni urban scene reconstruction," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
[66] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proc. Conf. Robot Learn. (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 78, 2017, pp. 1–16.
[67] R. Martín-Martín, M. Patel, H. Rezatofighi, A. Shenoi, J. Gwak, E. Frankel, A. Sadeghian, and S. Savarese, "JRDB: A Dataset and Benchmark of Egocentric Robot Visual Perception of Humans in Built Environments," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, pp. 6748–6765, 2023.

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